D-2 Water Level Adjustment (WLA)

The user must be able to adjust depth information by water level height in areas where both S-102 and S-104 data products are available.

1. The system must default to no water level adjustment.
2. The mariner must be able to select one of the following methods of depth adjustment:
   a. Current (or user selected) date and time;
   b. A mariner specified date and time period;
   c. Where the route includes a schedule, the predicted date and time of transit in each area along a route
3. When Water Level Adjustment is activated as defined in this section:
   a. The functionality and portrayal of the safety contour, depth zone shades, safety depth and indication of isolated dangers must use the adjusted depth;
   b. The ECDIS pick report must indicate both adjusted and unadjusted depth;
   c. Details of the Water Level Adjustment must be readily available, such as the data source, temporal extent, and applicable areas;
   d. It must be possible to de-activate Water Level Adjustment via simple operator action;
   e. There must be a permanent indication as described in D-2.5, D-2.6 and D-2.7.
4. The ECDIS voyage recording must include:
   a. The state of Water Level Adjustment (method applied);
   b. The mariner specified date and time or date and time period when in use for WLA, or the scheduled date and time, the check area distance and the time resolution when in use for WLA.

D-2.1 Scope

This section defines how the "**ECDIS Water Level Adjustment feature**" is implemented. Water Level Adjustment is referred to in this section as WLA. This section defines how the adjustment of depth information by water level is provided for:

- *Selected single date and time;*
- *Selected date and time period;*
- *Linked to an estimated route schedule with selected check distance and time resolution.*

D-2.2 Constraints on input data

1. Coverage – S-104 data is not expected to overlap, but in case of an overlap of greater than one grid cell[3] the approach must be same as for overlapping S-102 data. The ECDIS must provide a permanent indication of "Overlap" and the user **must** have the ability to select which producer in the overlapped area has priority and will be selected for processing WLA.
2. As with S-102, each S-104 point is assigned a rectangular extent with nearest neighbour interpolation.
3. WLA must not be applied unless the S-102 and S-104 features are on the same vertical datum. If the S-102 and S-104 features are not on the same vertical datum then an indication "Incompatible vertical datums" must be given to the user.

D-2.3 User inputs

When WLA is selected for use, in areas where WLA coverage exists, one of the three different options must be used:

1. WLA Option 1: WLA at a single date and time (current or user selected).

[3] In the case of a difference in resolution the maximum overlap is the size of the larger of the overlapping grid cell.

2. WLA Option 2: WLA for a date and time period (from start date and time to end date and time).
3. WLA Option 3: WLA linked to an estimated route schedule. In this case the user also specifies:
   a. A distance parameter, the limit of the check area as specified by IMO MSC 530(106) 11.3.5 when route planning and MSC.530(106)/Rev.1 11.4.4 when route monitoring.
   b. A time resolution t_u used to construct the individual WLA sections. This time resolution reflects the uncertainty or tolerance of the time schedule of the route, for example 10 minutes if the user assumes that they could follow the time schedule within 10 minutes.

D-2.4 Implementation - general

WLA can be applied only in areas where there is data from both S-102 and S-104. For example, the intersection of the Red and Blue outline in figure D-6 below. WLA is applied to S-101 features after substitution of depths by S-102 as defined in clauses D-1.2 and D-1-3.

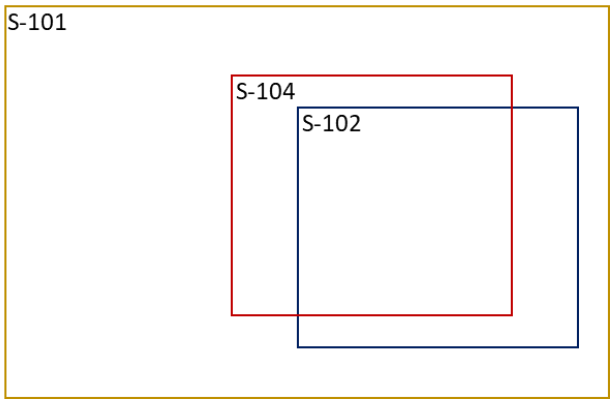


Figure D-6 - WLA can only be computed in areas where there is S-104 and S-102 data

Further, WLA can only be carried out:

1. When WLA option 1 or WLA option 2 is selected, where the temporal extent of the S-104 fully covers the required date and time instant or period selected by the user
2. When WLA option 3 is selected, where the S-104 temporal extent of the S-104 fully covers the estimated date and time of a part of the route; see figure D-2-16.

Where these conditions are not met the WLA processing must not be carried out.

When WLA has been processed the area for which it is defined must be surrounded by a boundary line using colour token DEPWL[4]; see Figure D-7.

[4] Thick dark blue dash line, colour token DEPWL xyL values are: DAY x= 0.15, y=¬0.15, L=30.0; DUSK x=0.15, y=0.15, L=7.5; NIGHT x=0.15, y=0.15, L=1.2.

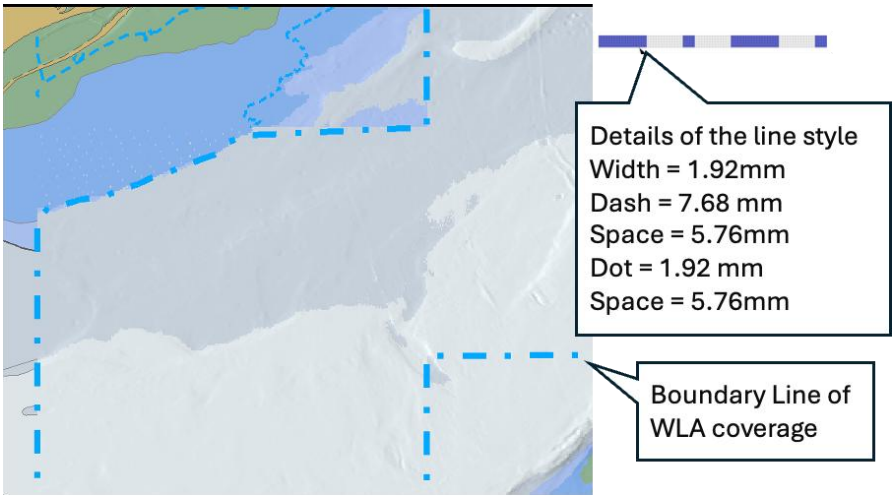


Figure D-7 – Boundary line of WLA coverage

NOTE: In this example S-102 coverage is larger than WLA coverage.

In addition to the display of the boundary line of the WLA coverage there should be a permanent indication about the application of WLA and the applied date and time, see details in WLA Options 1, 2 and 3.

OEMs are free to design their user interface. Usable ideas include, for example, a mouse roller to change datetime, use of a slider to change datetime, or even to provide an animation from user selected start date and time to user selected end date and time.

D-2.5 Implementation of WLA Option 1 – WLA for a single datetime instant

WLA is based on S-104 values closest to the selected datetime instant. Each S-102 point has an extent of coverage which is closest to it. Each S-104 point, similarly has an extent. The adjustment of the S-102 values is calculated by adjusting each S-102 point by the shoalest of the S-104 values, for all S-104 points whose extent intersects the extent of the S-102 point.

When an S-104 record does not exist for the precise time specified the shoalest of the two S-104 adjacent values is selected from the S-104 dataset. S-104 values can only be selected within the temporal extent of the S-104 dataset. In the example shown in Figure D-8 (where S-104 values, “V” defined every 15 minutes and a user selected datetime of 07:24) the two values selected are 07:00 and 07:30, WLA would select 1.3m (the 07:00 value).

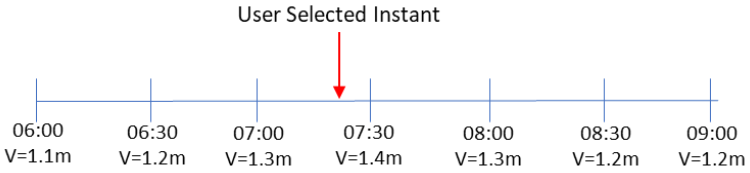


Figure D-8 – Selection of time-varying value

In Figure D-9 below, the S-102 point X is adjusted by the shoalest (that is, minimum) value of the S-104 values from (a), (b), (c) and (d) at the required datetime instant because the S-102 point extent overlaps the S-104 extents of a,b,c and d.

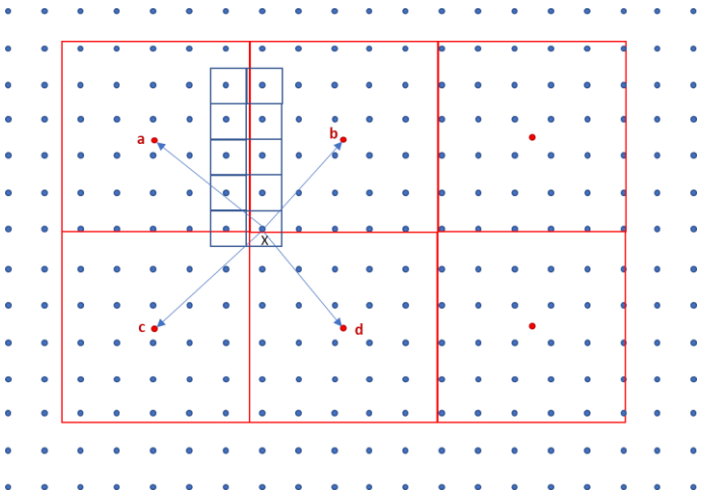


Figure D-9 – Adjustment of S-102 values by S-104

Format of the permanent indication is as below:

WLA 12:34 08 Nov 2021

D-2.6 Implementation of WLA Option 2 – WLA for a datetime range specified by the user as a time period

When WLA is based on a datetime range, then the process is identical to that followed for WLA Option 1 except each S-104 value selected is the shoalest of all values available in the S-104 within the selected time period.

All S-104 points contributing to the WLA must be defined across the time range required, otherwise the WLA is not computable and the user must be informed. The S-104 records selected are those which lie either within the user defined time period; or before the start point of the time period and just after the end point of the time period.

In the following examples:

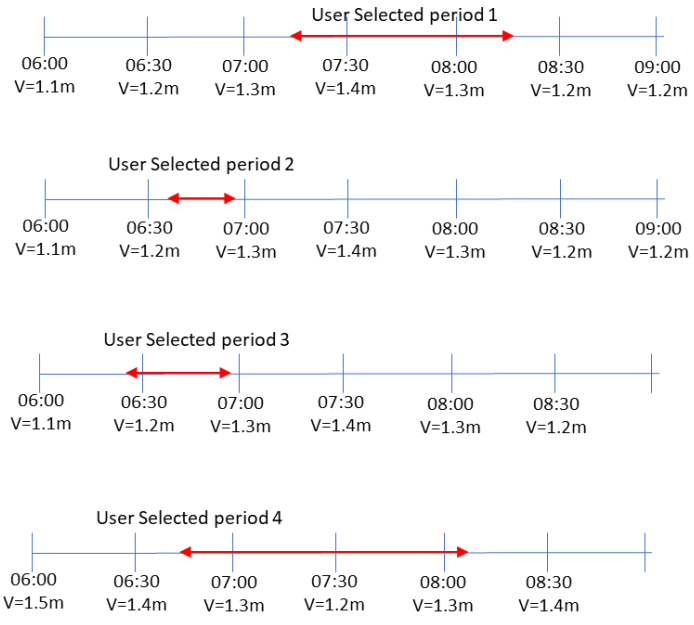


Figure D-10 – Examples of S-104 data selection

1. User selected period 1, four values are from 0700, 0730, 0800 and 0830 - WLA would select 08:30 value = 1.2m (even though the time period doesn't overlap).
2. User selected period 2, two values are from 0630 and 0700 - WLA would select 06:30 value V=1.2m (even though the time period doesn't overlap).

3. User selected period 3, three values are from 0600, 0630 and 0700 - WLA would select 06:00 V=1.1m (even though the time period doesn't overlap).
4. User selected period 4, five values are from 0630, 0700, 0730, 0800 and 0830 – WLA would select 07:30 V=1.2m.

When adjusting depth values the shoalest (smallest) value from the selected S-104 records must be used in order to produce the safest WLA values.

Format of the permanent indication is as below:

WLA from 12:34 08 Nov 2021 to 14:56 08 Nov 2021

D-2.7 Implementation of WLA Option 3 – linking of WLA to a defined route with planned waypoints and times

When WLA is based on a route then the limit of check area around the route is set by the user as specified by IMO MSC 530(106) 11.3.7.

The same user-specified distance must be used for the check of safety contour, prohibited areas, geographic areas for which special conditions exist and navigational hazards and equivalent requirements when route monitoring

The WLA is processed within this check area. The boundary of the area of the display where WLA has been carried out must be displayed (as shown in Figure D-16). In the process description this distance is referred to as ‘a’.

A route could be either a Planned route or a Monitored route. Both could be processed for the WLA, but not at the same time in a single display area. In case of multiple display areas, it is possible that one area is WLA processed for Planned route and another area is WLA processed for the Monitored route.

For a Planned route the datetime period applied for the WLA process is based on the schedule of the planned route itself.

For a Monitored route the datetime period applied for the WLA process is user selectable either:

1. Based on the planned schedule of the monitored route itself; or
2. Based on the monitored route adjusted for the current own ship position.

When WLA is based on the planned schedule and own ship is not keeping to schedule a Caution must be raised to indicate the water level being experienced may be different to that being applied by the ECDIS

The region of WLA is restricted to a distinguishable bordered polygon around the route. The process works as follows:

1. For each section of the route a series of estimated time markers is defined along the route at times t_0 – t_n. The diagram also shows the “limit of check area” as specified by IMO MSC 530(106) 11.3.5. Each time marker is delimited before and after by the time +/- half the user selected t_u interval so the rectangle in the diagram represents the extent of time t_1 which ranges from time $t_1 - \frac{t_u}{2}$ to $t_1 + \frac{t_u}{2}$. This is the WLA adjustment polygon (shown in red) corresponding to time t_1:

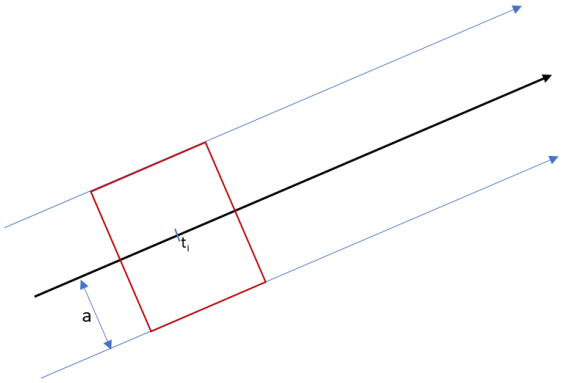


Figure D-11 - WLA adjustment polygon

2. Where the route follows a curved section the polygon formed is not a rectangle but an area defined by the user selected limit as shown by areas a and b:

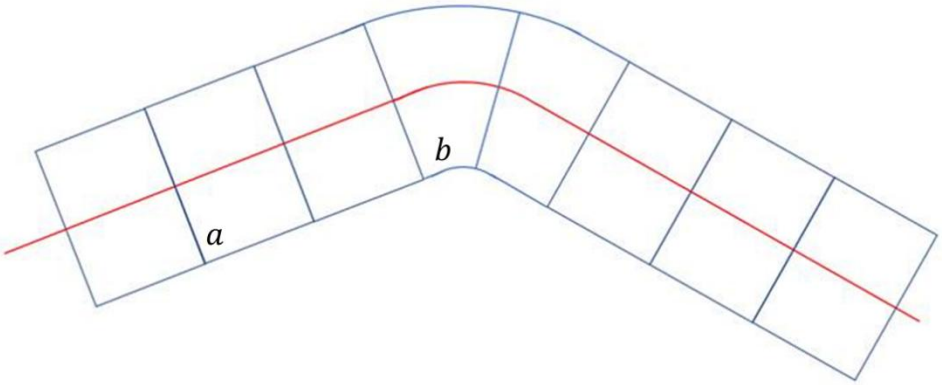


Figure D-12 – WLA adjustment polygon along curved section

3. The individual S-102 points (blue dots in Figure D-13 below) are assigned rectangular extents. For each WLA adjustment rectangle corresponding to each t_i the S-102 extents which spatially intersect are selected. A full example is shown in Figure D-13 for one of the WLA polygons (shown in red).

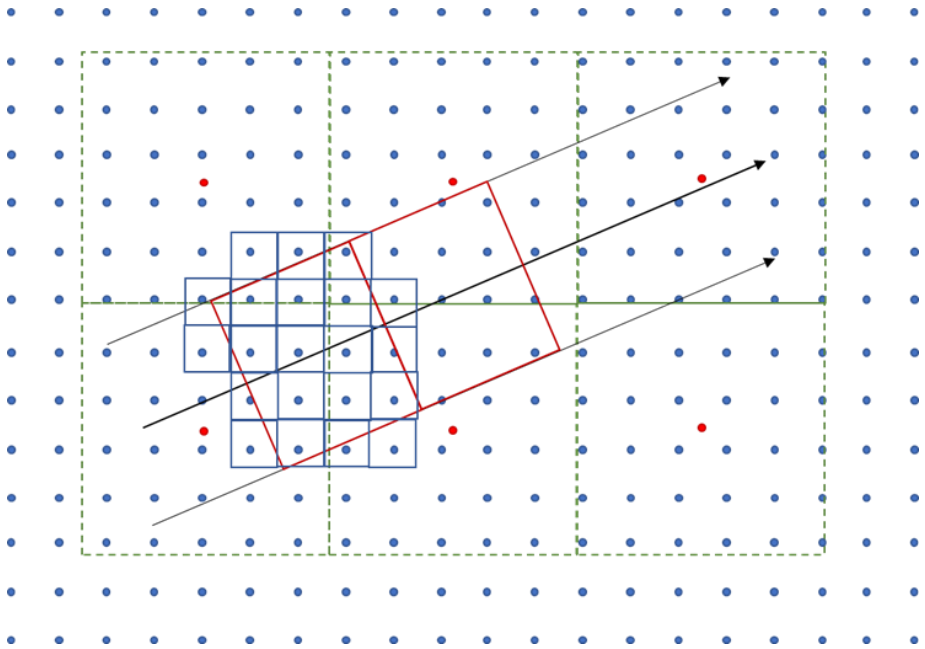


Figure D-13 – Grid cell selection

4. For each S-102 point whose extent intersects the WLA adjustment polygon the Water level is adjusted using a similar process to Option 1. The S-104 value used for adjustment is the shoalest value from each of the S-104 points whose extents intersect the S-102 point (as described in WLA Option 1) across the time period $(t_i - \frac{t_u}{2})$ and $(t_i + \frac{t_u}{2})$, In Figure D-14, illustrating this step, two S-102 points are adjusted by S-104 values in the time period $(t_1 - \frac{t_u}{2})$ and $(t_1 + \frac{t_u}{2})$, for calculation of t_1. S-102 point x is adjusted using values drawn from S-104 points ***a*** and ***b*** (because its extent intersects the extents of points ***a*** and ***b***) and S-102 point y is only adjusted with values from S-104 point ***b***.

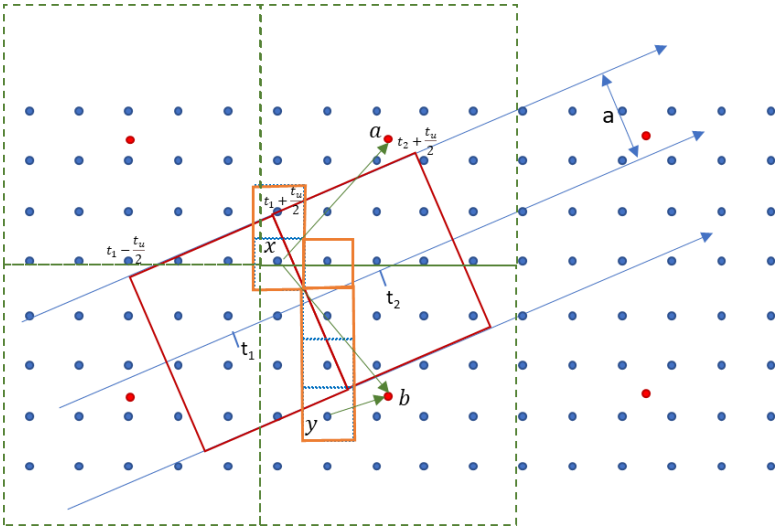


Figure D-14 – Selection of S-104 data

5. Each of the S-102 points are assigned the adjusted water level equal to the S-102 value + the calculated (shoalest) S-104 value as defined in the previous step.
6. Once the WLA polygon for time t_i is processed the WLA polygon for time t_{i+1} is processed.
7. S-102 points whose extents are intersected by the <u>both</u> the WLA polygon for t_i and t_{i+1} are assigned the shoaler of the two values, that is the shoalest of the S-104 values between the two time periods $(t_i - \frac{t_u}{2})$ and $(t_i + \frac{t_u}{2})$, and $(t_{i+1} - \frac{t_u}{2})$ and $(t_{i+1} + \frac{t_u}{2})$.

This completes the WLA process for (red) polygon t_i. In Figure D-15 below the orange border shows S-102 points lying in WLA polygons t_1 and t_2 and which would be assigned the shoaler of the two S-104 values corresponding to the t_1 and t_2 WLA process.

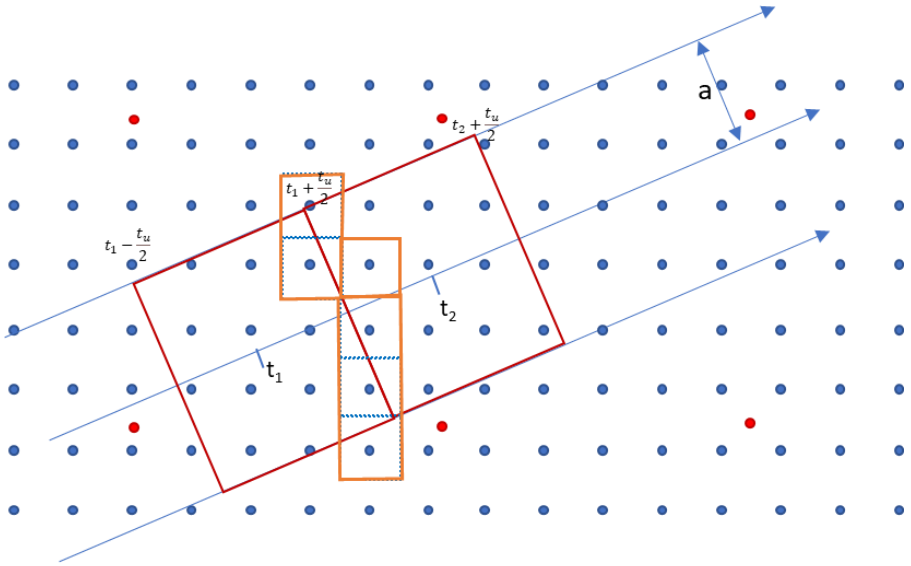


Figure D-15 – Time-dependent adjustment of S-102 data with S-104 data

This process is extended to all t_i along the planned route.

Figure D-16 shows a boundary line around all the WLA processed S-102 grid cells. This marks the boundary of the Water Level Adjustment area and requires portrayal to inform the user which areas of

the display are subject to WLA. This area is surrounded by a distinguishable boundary line using colour token DEPWL[5]).

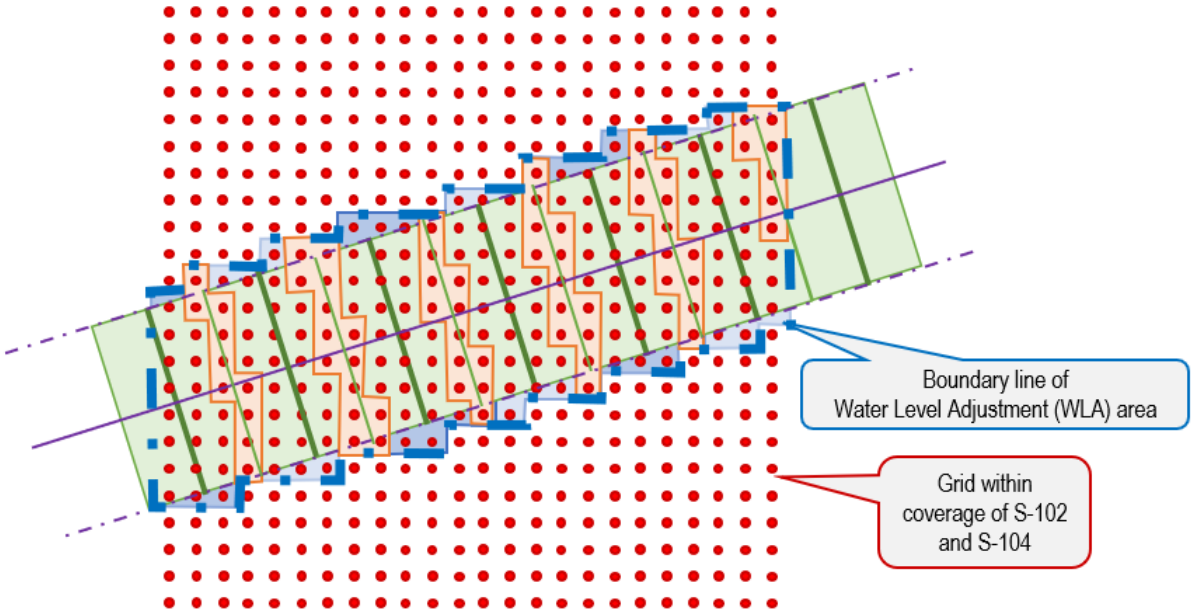


Figure D-16 – Boundary of adjusted area

Figure D-17 below shows how the rectangle is positioned in the middle of S-104 coverage area and at the edge of the S-104 coverage are.

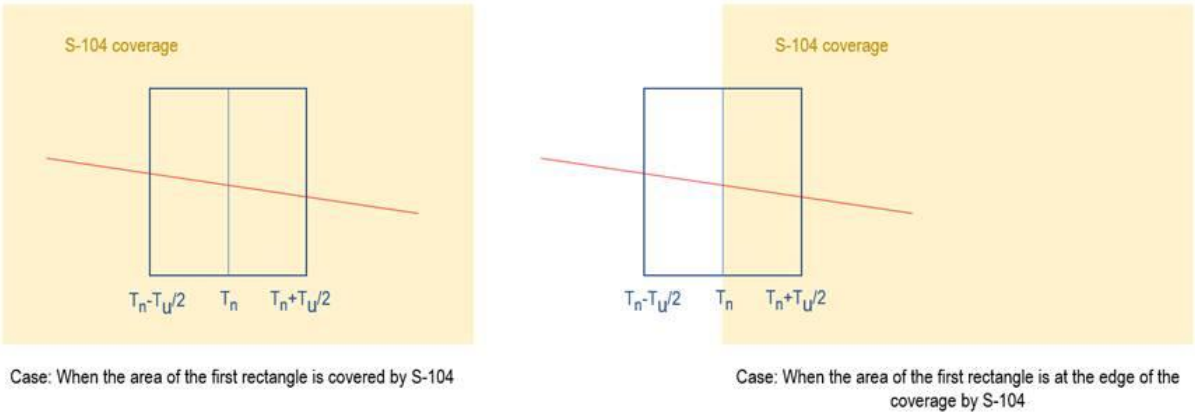


Figure D-17 – Positioning in interior and at edge

Format of the permanent indication is as below:

WLA from 12:34 08 Nov 2021 to 14:56 08 Nov 2021

[5] *thick dark blue dash dot line,* colour token DEPWL, xyL values are: DAY x= 0.15, y=¬0.15, L=30.0; DUSK x=0.15, y=0.15, L=7.5; NIGHT x=0.15, y=0.15, L=1.2

D-3 Treatment of Depth and Water Level Related S-101 Features

D-3.1 Substitution and adjustment of depth values

In areas covered by only S-102, all depth values must be substituted for all ENC features at all scales which have depth attribution (that is, the attribute *valueOfSounding* bound to them in the S-101 Feature Catalogue)

In areas where WLA is processed, all depth values must be adjusted for all ENC features at all scales which have depth attribution (that is, the attribute *valueOfSounding* bound to them in the S-101 Feature Catalogue).

D-3.2 Areas covered by S-102 only or by both S-102 and S-104

Adjustment of different geographic primitives.

1. Point Features
    - For Areas covered by S-102 only, a value for the attribute *valueOfSounding* must be taken from the S-102 grid cell extents which intersect the point feature (as shown in Figure D-18).
    - For Areas covered by S-102 and S-104, the value for attribute *valueOfSounding* must be taken from the S-102 grid cell extents which intersect the point feature. WLA is then applied to this value using S-104 records selected from the S-104 grid cell intersecting the point feature. The S-104 value selected is as defined in sections D-2.5, D-2.6 and D-2.7.
2. For Curve and Surface features
    - For Areas covered by S-102 only, the value for *valueOfSounding* must be the shoalest value of all S-102 grid cells whose extents intersect the feature's geometry within the S-102 coverage available. If the curve or surface feature is not completely within the S-102 then the value defined is the shoalest resulting from the original feature value in S-101 attribute and the value selected from the intersecting S-102 grid cells.
    - For Areas covered by S-102 and S-104, the value for *valueOfSounding* is first selected, as the shoalest value of all S-102 grid cells whose extents intersect the feature's geometry. WLA adjustment must then applied to this value by selecting the shoalest value from all intersecting S-104 grid cells. The value selected is as defined in clauses D-2.5, D-2.6 and D-2.7 and shown in Figure D-19. If the curve or surface feature is not completely within the S-104 coverage then the adjusted value is the shoalest value resulting from the original feature value in S-101 attribute and the WLA adjusted value.

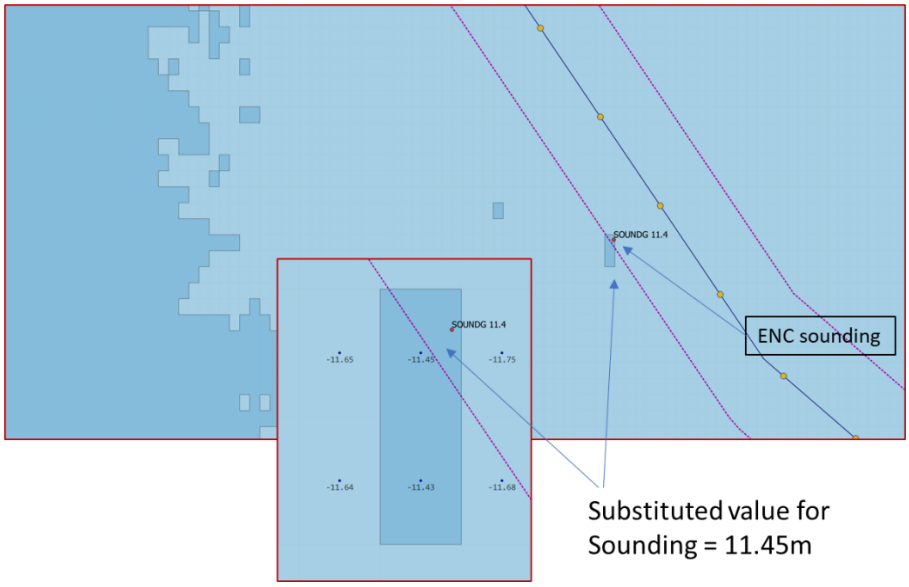


Figure D-18 – Substitution of sounding value from S-102

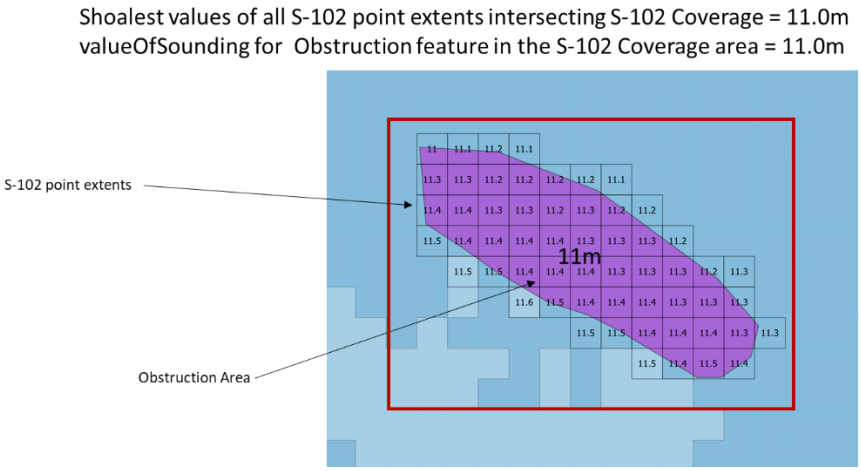


Figure D-19 - Substitution of *valueOfSounding* attribute in S-101 Obstruction from S-102 data

The S-101 features for which this depth substitution must be implemeneted within S-102 coverage areas are all those which bind *valueOfSounding* in the S-101 feature catalogue along with multipoint geometries, unless those specifically excluded in this section.

Soundings are included (either individual soundings or those which are part of an array) substituting the S-102 grid cell whose extents intersect the sounding position for the defined ZCOO (ISO8211), See figure D-3-18.

If the depth substituted S-101 feature is covered by S-104, then WLA must be processed using the same method as for the underlying depth area.

Depth values must also be substituted in all ENC features at all scales which have the attribute *depthRangeMinimumValue* bound to them in the S-101 Feature Catalogue.

For all features with substituted (and possibly adjusted values), the ECDIS Pick Report must indicate the substituted/adjusted value and its source. The format and portrayal of depths and drying heights (that is, number of decimals, etc) is unchanged.

For example, for S-102 only covered area:

Value Of Sounding 12.3 m [S-102]

When an S-101 attribute has been WLA adjusted the pick report must indicate the WLA adjusted value, source and time/date. The format of time date is: hh:mm dd mmm yyyy.

For example, for both S-102 and S-104 covered area:

Value Of Sounding 15.5m [WLA 12:34 08 Nov 2021]

Value Of Sounding 15.5m [WLA from 12:34 08 Nov 2021 to 14:56 08 Nov 2021]

D-3.3 Adjustment of heights and vertical clearance values

Any vertical measurement which is referenced to the same vertical datum as the S-104 data must be adjusted using the intersecting S-104 grid cell water level value. This must be applied on any features within the coverage of S-104. Where a feature is not completely within the S-104 coverage the adjusted value is the minimum value resulting from the original feature value in the S-101 attribute and the WLA adjusted value.

Adjustment is defined for all features which bind *verticalDatum* in the S-101 feature catalogue unless specifically excluded in this section.

The simple attributes adjusted are *height* and *verticalClearanceValue*. When adjusting height or clearance values the largest value from the selected S-104 records must be used in order to produce the safest values.

The user must be given a permanent indication that the Water Level (S-104) values have adjusted attributes in the data display. Where values are not adjusted due to an incompatibility of vertical datums in the data, the user must be given a similar permanent notification.

When either *height* or *verticalClearanceValue* in S-101 have been substituted or adjusted the ECDIS Pick Report must indicate the substituted value, source and time/date. The format of the height or

vertical clearance values (that is, number of decimals, etc) is unchanged.

For example:

Vertical Clearance Value 5.3 m Mean Sea Level [WLA 12:34 08 Nov 2021]

Value Of Vertical Clearance 15.5m Mean Sea Level [WLA from 12:34 08 Nov 2021 to 14:56 08 Nov 2021]

D-3.4 Alerts and indication details

Substituted or adjusted values in ENCs at the largest scale must be used as input to any alert/indication processing. The substituted values are portrayed as shown in Figure D-20 which shows the graphical highlight (red boundary and red transparent fill) for the intersected safety contour.

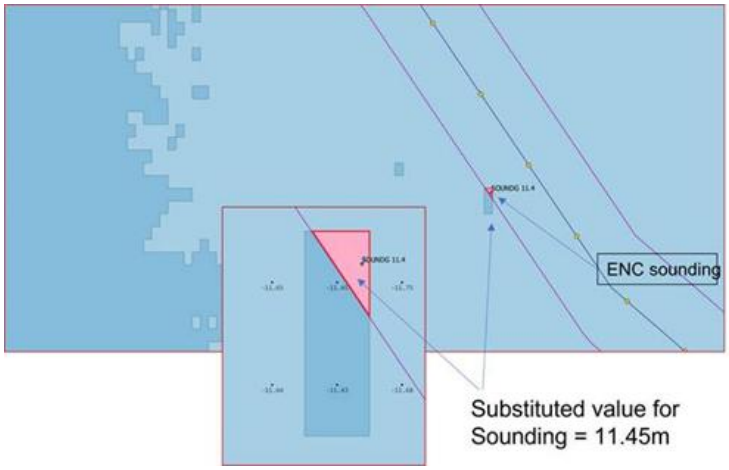


Figure D-20 – Graphical highlight for intersected safety contour

D-3.5 Legend details

The legend must additionally indicate the vertical datum of S-102 and S-104 If the vertical datums are the same then a single indication is enough to cover S-101, S-102 and S-104.